CSC4010: Parallel and Distributed Computing

OpenMP Continued

David Cutting
Email: d.cutting@qub.ac.uk — Web: davecutting.uk

1 Introduction

In this worksheet we will be continuing to look at OpenMP concepts and constructs.

Please note you will need to refer to the class examples (see Canvas for the repository links and also videos showing the demonstration) in addition to the step-by-step topics here.

You can choose your workflow but some examples will require you to ultimately run on Kelvin, some example job scripts are provided.

2 Programs

Download the zip file (to your local computer and/or to Kelvin via wget or transfer as you like) from:

`https://files.davecutting.uk/HPC/OpenMPContinuedPrograms.zip`

Inside the archive you will find a number of sub-folders described below, most containing a c file and some a job script for Kelvin. Where a job script is missing you may need to create one. See the comments in the programs for what you can try. Feel free to experiment with the code!

3 Challenges

First I'd suggest you revisit the introduction challenges and see if you can use the concepts we've seen since to improve them at all.

3.1 Reduction

Write an OpenMP program that programatically sets 8 threads in a parallel block and makes use of reduction.

Each thread should initialise a variable 'sum' to the respective thread ID and will compute a local sum, sum = sum + 10.

The reduced sum should be printed from outside the parallel block in the program.

3.2 Bubble Sort

Write a sequential program that initialises a large array of random integers (configurable how large but I'd go for 100,000+) and then performs a bubble sort. Keep track and output at the end (a) the loop count (how many times did the loop go through the array) and (b) the swap count (how many swaps were performed).

Parallelise this program checking performance in different configurations - what's the lowest wall time and best PE you can get? Are there any major issues that need worked around? How do you structure the parallel region(s)? Are there any limits to parallel efficiency?